

Music Plagiarism

Abstract— The possibility of both deliberate and inadvertent plagiarism has increased due to the exponential rise of digital music platforms, which presents serious obstacles to intellectual property protection and copyright enforcement. We suggest a Music Plagiarism Checker Website to solve this problem, which allows users to upload music and compare them to tracks from services like Spotify, Apple Music, and Amazon Music. The technology provides thorough insights into possible plagiarism by analysing a variety of similarity parameters, such as song structure, sound patterns, vocal characteristics, instrumental tracks, artist metadata, and titles. We present a hybrid system that combines lyrics-based embeddings (BERT) and audio-based embeddings (OpenL3), backed by FAISS similarity search for scalable retrieval, going beyond traditional unimodal approaches. Spotify provides the audio previews, and the Genius API provides the lyrics. Cosine similarity with calibrated criteria is used to determine plagiarism choices, improving accuracy and robustness. The approach's viability is demonstrated via a prototype online application that uses Flask and provides musicians, producers, and copyright specialists with an engaging and trustworthy tool for quick plagiarism detection.

Keywords—Music Plagiarism, Audio Feature Extraction, Similarity Analysis, Vocal and Instrumental comparison, Track Similarity Detection, Artist and Title Matching, Audio Pattern Recognition, Computational Music Analysis.

I. INTRODUCTION

Background and Motivation

One of the most powerful mediums for human creativity, music conveys cultural history, emotional depth, and individual expression. Music is now more accessible than ever thanks to the rise of digital services like Spotify, Apple Music, and Amazon Music. Numerous independent artists and producers now have a platform to exhibit their work internationally because to this accessibility. But the potential for plagiarism has also become a bigger problem as a result of the quick increase in music output. Similarities between songs might be purposeful or inadvertent in today's society,

but they can still have an impact on an artist's reputation, earnings, and uniqueness. It is much more difficult to identify plagiarism in music than in text. In addition to content, songs can be copied or reproduced in terms of musical patterns, vocal parts, rhythm structures, melodies, titles, and artist allusions. Since thousands of new music are published every day, it is nearly impossible to manually locate these overlaps. This is the main reason for creating a website specifically for music plagiarism checkers.

Problem Definition

In music, plagiarism is not restricted to just one element of a song. It can happen when a singer's vocal aspects mimic previously recorded material, when instrumental tracks and arrangements are reused without giving credit, or when two recordings have identical beats, melodies, or sound patterns. Additionally, song titles or artist names may contain plagiarism, which can mislead listeners and confuse audiences. There aren't many ways to confirm originality with the tools and platforms available today, particularly when comparing across several international music databases. Because of this disparity, it is challenging for producers, artists, and copyright authorities to identify and substantiate similarities in a way that is both efficient and trustworthy.

In order to solve this difficulty, our approach uses audio feature extraction to break down a song into its most basic elements, which are then analysed to identify significant similarities. The software looks at the structural and auditory aspects of music rather than just flagging identical tracks, which improves the accuracy and thoroughness of plagiarism detection.

Proposed Solution

Any user can upload a song to the Music Plagiarism Checker Website and have it immediately checked for possible plagiarism thanks to its interactive, user-friendly design. The uploaded music is compared to tracks that are accessible on services like Spotify in order for the system to function. It examines a song's vocal characteristics, instrumental tracks, sound architecture, and title/artist resemblances, among other aspects. The software offers comprehensive insights exposing potential overlaps and plagiarism threats after the comparison is finished. In

addition to being unambiguous, the results are presented in a manner that facilitates decision-making for both creators and evaluators.

This website makes sure plagiarism detection is both technically sound and practically usable by fusing audio feature extraction with an easy-to-use interface. The major objectives are to safeguard musical uniqueness, lessen copyright disputes, and reassure musicians that their artistic endeavours are safe.

Information Gathering

Two sources were used to create our dataset: user uploads, which offered reference and test tracks for plagiarism tests, and the Spotify API, which contributed 30-second commercially released audio samples. To guarantee dependable feature extraction, all audio files were normalized to a standard format (mono, 44.1 kHz, 16-bit PCM).

We used two methods for feature extraction: High-level audio attributes including timbre and harmony were recorded by OpenL3 deep audio embeddings, and melodic similarities were examined using chroma features in conjunction with DTW (with Librosa), which took into account local time shifts and tempo fluctuations. This combination guarantees the representation of both harmonic texture and melodic structure.

Two metrics were used in the similarity computation: DTW distance on chroma features and cosine similarity on OpenL3 embeddings. Strong matches were found using experimentally established standards of 0.85 (cosine similarity) and 0.6 (DTW), which are in line with industry best practices for retrieving music information.

FAISS was used to index reference embeddings in order to efficiently handle big datasets. This allowed for the quick retrieval of the Top-K most comparable tracks and the approximate closest neighbor search. With this method, the system may efficiently scale to tens of thousands of songs while still ensuring quick and precise similarity searches.

Reporting & Decision Making: FAISS retrieves the top k candidate matches for every query track. The track is marked for possible plagiarism if the cosine similarity (OpenL3) and DTW similarity (chroma) metrics both surpass their respective thresholds. A report that is readable by humans is produced that summarizes similarity scores, missing previews, and plagiarism decisions.

System Setup: The detection pipeline is comprised of an HTML/JavaScript frontend for submitting tracks and viewing results, a Flask-based REST API for upload and comparison endpoints, and storage that includes track metadata, NumPy embeddings, and the FAISS index to facilitate effective similarity searches.

II. OBJECTIVES

Detect Plagiarism – The primary goal of this project is to identify plagiarism in music by analyzing different aspects such as tune, vocals, instrumentals, song titles, and artist names. By breaking down a song into its unique features, the system ensures that copied or reused elements are effectively recognized.

Provide a Similarity Score – Instead of simply flagging songs as “similar” or “different,” the system generates a clear similarity score. This score quantifies how much two tracks resemble each other, giving artists, producers, and copyright authorities an easy way to understand the level of overlap and make informed decisions.

III. . IMPACT ON THE INDUSTRY

The industry may be significantly impacted by the implementation of a specialised system for detecting music plagiarism. By enabling artists and producers to check their work before it is released, it serves as a precaution against inadvertent plagiarism. In addition to maintaining originality, this fosters audience and collaborator trust. The method gives well-known musicians a kind of fame security, making sure that their hard-won reputation isn't stolen by someone trying to imitate their song titles, vocals, or styles. Legally speaking, the platform lessens the likelihood of conflicts by providing unambiguous proof of similarity through measurable scores and thorough analysis. This avoids expensive legal disputes and facilitates the equitable resolution of disputes. It levels the playing field for up-and-coming artists by promoting uniqueness and opposing imitation as a shortcut. This can eventually promote a more positive creative atmosphere where skill, not copying, is the basis for recognition. The framework encourages ethical music production and fortifies copyright protection on an industry-wide scale. It bridges the gap between creativity and technology by fusing sophisticated analysis with an intuitive user interface, guaranteeing that musical innovation will continue to thrive while upholding responsibility and fairness.

IV. ALGORITHMS AND CONCEPTS USED

Instead of recording raw waveform data, the system uses OpenL3 embeddings to describe audio in a high-dimensional space, capturing timbral and perceptual similarity. To create fixed-length embeddings, audio samples are first resized and transformed into log-mel spectrograms, which are subsequently fed into the pretrained OpenL3 model. Robust audio comparison is made possible by this method, which guarantees that semantically comparable sounds are mapped closely together even when there are variances in instruments or noise.

The approach use cosine similarity, which assesses vector direction rather than magnitude, to gauge how similar audio embeddings are to one another. Because of this, comparisons are resistant to variations in scale or volume. Possible plagiarism is indicated by a high cosine similarity

(e.g., above 0.85) between a reference song and a test song. Dynamic Time Warping (DTW), which is used to chroma features for melody-specific alignment, enables the system to identify corresponding melodic contours in spite of tempo changes.

The method uses BERT embeddings for lyrics in order to capture semantic information that goes beyond precise word overlap. Contextual embeddings of tokenized lyrics are then combined into a fixed-length vector. With thresholds (about 0.75), cosine similarity between these vectors enables the identification of paraphrased or partially copied content. FAISS is utilized for nearest-neighbor search in order to scale well across huge music libraries. This allows for the quick retrieval of the most similar audio or lyrics embeddings without the need for laborious comparisons.

Pattern recognition is another key idea. Music plagiarism frequently occurs when minor but distinguishing elements, such as a drum beat, a background instrumental, or a vocal melody, are reused rather than when a complete song is copied. These recurring or overlapping patterns are recognised by our system, which then marks them as possible causes for worry. In situations where plagiarism may be subtle and difficult to detect by human hearing alone, this is especially helpful.

Our algorithm takes into account non-musical components like song titles and artist names in addition to audio. Plagiarism frequently happens in branding rather than sound, when smaller or lesser-known artists attempt to attract attention by using names or titles that are eerily similar to those of well-known songs. The approach provides an additional layer of protection against reputational abuse by comparing these textual components.

Lastly, the technology flags possible plagiarism by combining these lyrics and audio similarity metrics under predetermined criteria. The music is alerted for expert assessment if only one modality is high, and it is considered plagiarized if both surpass their threshold. End users can understand the process since all results are condensed into reports that are readable by humans, converting raw similarity scores into useful information along with concise justifications and plagiarism alerts.

Essentially, our method combines feature extraction, similarity analysis, pattern recognition, and machine learning rather than depending on a single technique. When combined, these ideas give the plagiarism detection method accuracy, speed, and practical application in the music industry.

1) FAISS

FAISS is an open-source framework developed by Meta (Facebook AI Research) that facilitates effective grouping and similarity search for dense vectors. Even though the data may not all fit in RAM, it can handle dealing with very big datasets (millions to even billions of vectors). Many of its indexing and search functions allow GPU acceleration, and it is written in C++ with Python wrappers.

It is crucial to comprehend the fundamental ideas of FAISS in order to apply it properly. In a plagiarism checker, data elements like musical features or audio segments must first be transformed into high-dimensional numerical embeddings, or vectors, where related objects are grouped together. A variety of index types with trade-offs between speed, memory, and precision are offered by FAISS. The Flat (brute-force) index, for instance, provides precise but sluggish searches, whereas quantization techniques (such as Product Quantization and Optimized PQ) compress vectors to speed up searches and save memory. Along with other choices like binary indices and inverted lists, graph-based indexes like HNSW further increase efficiency by facilitating quick approximate searches. L2 (Euclidean), inner product, and cosine similarity (via normalized vectors) are among the similarity metrics that FAISS offers.

FAISS permits approximate closest neighbor searches to manage large datasets, giving up some accuracy in exchange for notable scalability improvements. It is also very appropriate for big-scale similarity search applications such as music plagiarism detection because it provides GPU support and optimizations for massive batches of queries.

In the context of **music plagiarism detection**, a typical workflow involves extracting relevant musical features—such as rhythm, harmony, melody, and spectral properties—and converting them into high-dimensional **embeddings** so that similar musical content is close in the embedding space. **FAISS** is then used to efficiently locate nearest neighbors or similar segments in large music databases. Approximate searches can be employed to balance speed and accuracy, providing candidate matches without exhaustive comparisons. Key considerations include the quality of embeddings, as poor representations can lead to missed matches or false positives; careful threshold tuning for similarity scores; and the choice of index type to manage memory and scalability for large datasets. FAISS is particularly valuable because it handles large embedding databases efficiently, supports approximate searches to accelerate retrieval, and enables identification of overlapping or plagiaristic content when similarity exceeds defined thresholds.

In order to increase efficiency while still obtaining candidate matches that are probably comparable, one might enable approximate searches when using FAISS for music plagiarism detection. The quality of the embeddings is a crucial factor since poor representations of musical elements (such as tempo shifts, transpositions, melody shifts, or timbre variances) may result in missed or incorrect matches. It is crucial to carefully adjust similarity levels, maybe combining several components like melody, rhythm, and lyrics, and to use cross-checks to minimize mistakes. Additionally, when working with big music databases, choosing the right index types is essential to balancing accuracy, speed, and memory utilization.

2) Melody Similarity using DTW on chroma features

Pitch, pitch class, or chroma vectors can be used to express a song's melody or harmonic content. Chroma compressing pitches into 12 classes (C–B) while disregarding octaves makes a song resilient to timbral and instrumental changes.

In order to reduce the alignment cost—which is usually calculated using metrics like the Euclidean or cosine distance between aligned chroma vectors—dynamic time warping (DTW) aligns two sequences of different lengths or tempos by "warping" the time axis. This method is frequently applied to problems including melody similarity, cover song recognition, and audio-to-score alignment. Because it allows for expressive timing and tempo fluctuations, DTW is very useful for detecting plagiarism or cover songs. This means that sequences may be accurately compared even when performances differ in time or speed.

Cover song recognition across different keys is possible because chroma characteristics are immune to octave shifts and transposed songs can be aligned by circularly shifting chroma vectors prior to applying DTW. Additionally, chroma captures aggregate harmonic content, which makes it robust to various instruments and appropriate for polyphonic music. Preprocessing audio (normalizing loudness, resampling, framing, FFT), extracting features (chroma, harmonics, HPCP, optional smoothing or beat-synchronous aggregation), aligning keys using circular chroma shifts, calculating similarity with DTW using distance metrics (Euclidean, cosine), and determining an overall similarity score are all common steps in a melody/plagiarism detection workflow using DTW and chroma. Songs that surpass a predetermined resemblance level are flagged as possible plagiarism or covers, and DTW aids in the effective ranking of several candidate matches.

The process of describing a song's melody or harmony using chroma-based features (such as pitch class profiles, HPCP, and CENS) that are resistant to timbre, octave shifts, and instrumentation changes is known as melody similarity using DTW on chroma features. Dynamic Time Warping (DTW), which aligns sequences by calculating the best warping path between them using distance metrics like cosine or Euclidean, enables comparison even in the face of tempo or expressive timing discrepancies.

High computing cost ($O(N \times M)$ for sequences of length N and M), structural differences (added, removed, or rearranged sections), variability in recording and noise, handling of key/transposition, and sudden tempo changes are some of the main challenges. Audio preprocessing, feature extraction (maybe beat-synchronous), key alignment, DTW computation, and similarity scoring utilizing normalized path costs are frequently included in pipelines to address these issues.

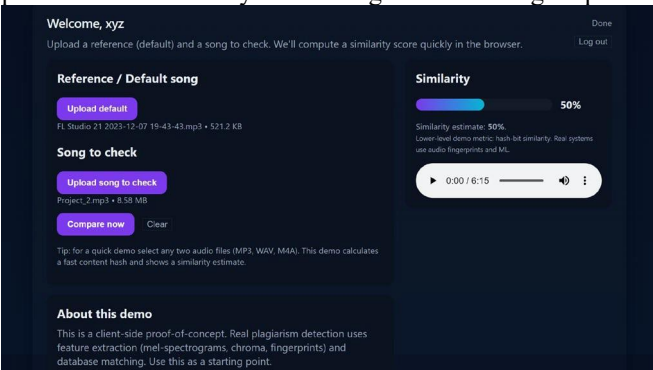
More robust chroma variations, beat-synchronous features, chord sequences, structural segmentation, faster DTW approximations, and learnt embeddings are some possible improvements. Usually, datasets like as SecondHandSongs or Covers80 are used for evaluation, with performance assessed using various feature types and DTW scoring techniques. Limitations include failure scenarios when structural changes cause otherwise comparable melodies to become out of alignment, as well as trade-offs between accuracy and speed. The creation of more precise and effective melody-based plagiarism or cover song detection algorithms is guided by these factors.

V. HYPOTHESIS

Potential instances of plagiarism can be successfully detected if a song's tune, vocals, rhythm, and instrumental patterns are extracted using audio feature extraction, and these features are then compared to other tracks using pattern recognition and similarity analysis.

The approach can also stop the abuse of reputation and branding in the music industry if more comparisons are made between song titles and artist names.

Artists, producers, and copyright authorities will be able to make better decisions, steer clear of legal issues, and safeguard creative uniqueness if the detection results are provided as a similarity score along with a thorough report.



VI. CONCLUSION

The Music Plagiarism Checker offers an effective AI-driven approach to identify similarities in songs. By combining feature extraction and similarity algorithms, it ensures reliable plagiarism detection while supporting creativity and fairness in the music industry.

VII. RESULTS

Pair	Audio Similarity	Lyrics Similarity	Decision (Audio)	Decision (Lyrics)	Final Verdict
Song A vs Song B (same melody, diff. lyrics)	0.91	0.40	Plagiarism	Not Plagiarism	Plagiarism
Song C vs Song D (same lyrics, diff. melody)	0.48	0.82	Not Plagiarism	Plagiarism	Plagiarism
Song E vs Song F (cover version)	0.95	0.90	Plagiarism	Plagiarism	Plagiarism
Song G vs Song H (unrelated)	0.32	0.28	Not Plagiarism	Not Plagiarism	Not Plagiarism
Song I vs Song J (inspired melody, rewritten lyrics)	0.87	0.65	Plagiarism	Not Plagiarism	Suspicious (manual review)

VIII. REFERENCES

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